TUTORIAL 3

Fields and Finite Fields

MOTIVATION AND GUIDING IDEA

Objectives. We will strengthen proofs by induction and proofs of equivalence, identify the field axioms, and calculate in finite fields. We will distinguish familiar real arithmetic from arithmetic adapted to the specified field.

The first part is a focused review of proof methods and the language of sets. The second part introduces fields and finite field arithmetic; the order used in class will follow the progress of the lectures.

MOTIVATION AND GUIDING IDEA

A proof must work within the given setting: equality of sets is proved by two inclusions, and a field identity is proved from the field axioms. In a finite field, addition, subtraction, multiplication, and division by nonzero elements are permitted; numerical notation does not imply an ordering like that of $\mathbb{R}$.

NOTATION AND CONVENTIONS

The symbol $x \in A$ expresses membership of an element, whereas $A \subseteq B$ expresses inclusion of sets. The quantifiers $\forall$ and $\exists$ mean ``for every'' and ``there exists''; their order matters. The set $A \setminus B$ consists of elements of A absent from B, and $|A|$ is its cardinality when A is finite. For a field F, write $F^\times = F \setminus \{0\}$. The notation $a \equiv b \pmod{p}$ means $p \mid (a-b)$. For prime p, writing $\mathbb{Z}_p = \{0, \dots, p-1\}$ uses representatives of residue classes. The symbol $\mathbb{F}_q$ denotes a field with q elements; when $q = p^n$ with $n > 1$, it must not be identified with arithmetic modulo q.

DEFINITION 3.1

Equality of sets and cancellation. For sets A, B, we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. To prove equality, take an arbitrary element from each side and prove that it belongs to the other side. In a field F, for $a, b, c \in F$ with $c \neq 0$,

$$ac = bc \quad \Longleftrightarrow \quad a = b,$$

because multiplication by c^{-1} is permitted. The condition $c \neq 0$ is essential. The characteristic of a field is the least positive integer p for which $p \cdot 1 = 0$, if such an integer exists; otherwise its characteristic is 0. The characteristic of a finite field is prime.

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3.1 Algebraic structure: fields

3.1.1 The field axioms

RECALL AND TOOLS

A field is a set F equipped with two operations, addition and multiplication. For all $a, b, c \in F$, the operations satisfy the following axioms. In particular, $0_{\mathbb{F}} \neq 1_{\mathbb{F}}$ is required:

- **Closure:** if $a, b \in F$, then $a +_{\mathbb{F}} b \in F$ and $a \times_{\mathbb{F}} b \in F$.
- **Commutativity:** $a +_{\mathbb{F}} b = b +_{\mathbb{F}} a$ and $a \times_{\mathbb{F}} b = b \times_{\mathbb{F}} a$.
- **Associativity:** $(a +_{\mathbb{F}} b) +_{\mathbb{F}} c = a +_{\mathbb{F}} (b +_{\mathbb{F}} c)$ and $(a \times_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} (b \times_{\mathbb{F}} c)$.
- **Distributivity:** $a \times_{\mathbb{F}} (b +_{\mathbb{F}} c) = a \times_{\mathbb{F}} b +_{\mathbb{F}} a \times_{\mathbb{F}} c$ and $(a +_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} c +_{\mathbb{F}} b \times_{\mathbb{F}} c$.
- **Identity elements:** there is $0_{\mathbb{F}} \in F$ such that $a +_{\mathbb{F}} 0_{\mathbb{F}} = a$ for every $a \in F$, and there is $1_{\mathbb{F}} \in F$ such that $a \times_{\mathbb{F}} 1_{\mathbb{F}} = a$ for every $a \in F$.
- **Additive and multiplicative inverses:** every $a \in F$ has an additive inverse $-a$ with $a +_{\mathbb{F}} (-a) = 0_{\mathbb{F}}$; every nonzero $a \in F$ has a multiplicative inverse a^{-1} with $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

EXAMPLE 3.1

- **Examples:** $\mathbb{Q}$ (rational numbers), $\mathbb{R}$ (real numbers), and $\mathbb{C}$ (complex numbers) are fields with their usual addition and multiplication.
- **Nonexamples:** $\mathbb{Z}$ and $\mathbb{N}$ are not fields, because most of their elements have no multiplicative inverse; for example, 3 has no inverse in $\mathbb{Z}$.

3.1.2 Notation

A general field is usually denoted by F (sometimes $\mathbb{F}$). When no particular field is intended, we write ``Let F be a field.''
Notation convention: when the field is clear from context, we omit its subscript from the operations and the elements $0, 1$.

EXERCISE 3.1

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that $(a^{-1})^{-1} = a$.

WORKED SOLUTION

Recall that every invertible field element has a unique multiplicative inverse, and in particular $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$. Put $x = (a^{-1})^{-1}$ and calculate:

$$(a^{-1}) \times_{\mathbb{F}} x = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a$$

$$a \times_{\mathbb{F}} (a^{-1}) \times_{\mathbb{F}} x = a \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} x = a \qquad (*)$$

Thus $x = a$, that is, $(a^{-1})^{-1} = a$.

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EXERCISE 3.1 CONTINUED

Remark: step $(*)$ uses $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

V-I-C-T-O-R-Y!

EXERCISE 3.2

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that the multiplicative inverse of a is unique.

WORKED SOLUTION

Suppose $b \in F$ satisfies $a \times_{\mathbb{F}} b = 1_{\mathbb{F}}$. Multiply on the left by a^{-1}:

$$a \times_{\mathbb{F}} b = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a^{-1}$$

$$(a^{-1}) \times_{\mathbb{F}} a \times_{\mathbb{F}} b = a^{-1} \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} b = a^{-1} \qquad (*)$$

Hence $b = a^{-1}$, so the inverse of a is unique. *Remark:* again, step $(*)$ uses $(a^{-1}) \times_{\mathbb{F}} a = 1_{\mathbb{F}}$.

🏆 **Victory!**

3.2 Further exercises on fields

EXERCISE 3.3

Let a, b be elements of a field $\mathbb{F}$. Please prove that if $a \times_{\mathbb{F}} b = 0_{\mathbb{F}}$, then $a = 0_{\mathbb{F}}$ or $b = 0_{\mathbb{F}}$.

WORKED SOLUTION

We distinguish two cases according to b. **Case 1:** $b = 0_{\mathbb{F}}$, and the conclusion already holds. **Case 2:** $b \neq 0_{\mathbb{F}}$. Since $\mathbb{F}$ is a field, b has an inverse b^{-1}. Multiplying both sides by b^{-1} gives

$$a \times_{\mathbb{F}} b \times_{\mathbb{F}} b^{-1} = 0_{\mathbb{F}} \times_{\mathbb{F}} b^{-1} \Longrightarrow a \times_{\mathbb{F}} 1_{\mathbb{F}} = a = 0_{\mathbb{F}}$$

And we're done. Victory!

PROPOSITION 3.1

A field has no nontrivial zero divisors.

EXERCISE 3.4

Let F be a field. On F^3, define

$$(a, b, c) \oplus (d, e, f) = (a + d, b + e, c + f),$$

$$(a, b, c) \otimes (d, e, f) = (ad, be, cf).$$

Please determine whether this is a field.

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EXERCISE 3.4 CONTINUED

WORKED SOLUTION

The zero element is $(0,0,0)$, but

$$(1,0,0)\otimes(0,1,0)=(0,0,0).$$

Neither factor is zero, since $1\neq 0$ in F. A field has no nontrivial zero divisors, so this structure is not a field.

You know what that means? V-I-C-T-O-R-Y!

EXAMPLE 3.2

In $\mathbb{Z}_4=\{0,1,2,3\}$ with addition and multiplication modulo 4, we have $2\cdot 2=0$ although $2\neq 0$. Thus 2 has no inverse: if $2b=1$, then $2=2(2b)=(2\cdot 2)b=0$, a contradiction. Hence $\mathbb{Z}_4$ is not a field. Complex arithmetic, polar form, and Euler notation are covered in Tutorial 1.

3.3 Congruence and Modular Arithmetic

DEFINITION 3.2

Write $\mathbb{Z}$ for the integers. For an integer $m\geq 2$, the notation

$$a\equiv b \pmod{m}$$

means that $a-b$ is divisible by m: there is $k\in\mathbb{Z}$ such that $a-b=km$. The residue class of a is the set

$$\overline{a}=\{a+km : k\in\mathbb{Z}\}.$$

Any two elements in the same class are congruent modulo m. There are exactly m classes, represented by $0,1,\ldots,m-1$.

EXAMPLE 3.3

Modulo 13, we have $-5\equiv 8$ and $34\equiv 8$. For example,

$$(-5)+34\equiv 8+8\equiv 3, \qquad (-5)\cdot 34\equiv 8\cdot 8\equiv 12 \pmod{13}.$$

PROPOSITION 3.2

Addition and multiplication of classes can be computed using representatives:

$$\overline{a}+\overline{b}=\overline{a+b}, \qquad \overline{a}\,\overline{b}=\overline{ab}.$$

The results do not depend on the representatives. Indeed, if $a'=a+km$ and $b'=b+\ell m$, then

$$(a'+b')-(a+b)=(k+\ell)m,$$

$$a'b'-ab=m(kb+\ell a+k\ell m).$$

Both differences are divisible by m.

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NOTATION AND CONVENTIONS

For a prime p, the classes with these operations form the field $\mathbb{Z}_p$, also denoted $\mathbb{F}_p$. We normally omit the bars and use the representatives $0, 1, \ldots, p-1$. Addition and multiplication are modulo p. Division by $a \neq 0$ means multiplication by its inverse a^{-1}, which satisfies $aa^{-1} = 1$ in the field.

PROPOSITION 3.3

Every nonzero $a \in \mathbb{Z}_p$ has an inverse. A prime dividing a product divides at least one factor. Thus

$$ax = ay \quad \Longrightarrow \quad a(x-y) = 0 \quad \Longrightarrow \quad x = y.$$

Multiplication by a therefore sends the $p-1$ nonzero elements to $p-1$ distinct nonzero elements. One of them is 1, so there is x satisfying $ax = 1$. The same cancellation argument gives uniqueness. The other field axioms follow from the laws of integer arithmetic.

NOTE

For example, $5^{-1} = 8$ in $\mathbb{Z}_{13}$ because $5 \cdot 8 = 40 \equiv 1$. Hence $5x = 7$ gives $x = 8 \cdot 7 = 56 \equiv 4$. Primality matters: modulo 4, we have $2 \cdot 2 = 0$ although neither factor is zero. These operations do not form a field. There are also finite fields whose number of elements is not prime; they are not obtained simply by integer arithmetic modulo that number.

EXERCISE 3.5

Let p be an odd prime. Please prove that every $t \in \mathbb{Z}_p$ can be written as $t = x^2 + y^2$. Please find such an expression for 7 in $\mathbb{Z}_{13}$.

WORKED SOLUTION

Let $Q = \{x^2 : x \in \mathbb{Z}_p\}$ be the set of squares. The notation $|Q|$ is the number of its distinct elements. If $x^2 = y^2$, then

$$(x-y)(x+y) = 0.$$

A field has no zero divisors, so $x = y$ or $x = -y$. For $x \neq 0$, the elements $x, -x$ are distinct: equality would give $2x = 0$, and odd p would imply $x = 0$. Thus the $p-1$ nonzero elements give exactly $(p-1)/2$ distinct squares. Including 0 gives

$$|Q| = \frac{p+1}{2}.$$

The set $t - Q = \{t - q : q \in Q\}$ has the same number of elements. If Q and $t - Q$ were disjoint, their union would contain $p+1$ elements inside a set of size p, a contradiction. Their intersection therefore contains an element: there are x, y such that $x^2 = t - y^2$, giving $t = x^2 + y^2$. In the example,

$$4^2 + 2^2 = 16 + 4 = 20 \equiv 7 \pmod{13}.$$

Another one down. Victory!

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EXERCISE 3.6

Let p be an odd prime, and define $h(x) = x + x^{-1}$ for $x \in \mathbb{Z}_p \setminus \{0\}$. Please prove that h takes exactly $(p+1)/2$ distinct values. Please find the set of values when $p = 11$.

WORKED SOLUTION

For nonzero x, y,

$$h(x) - h(y) = x - y + \frac{y - x}{xy} = \frac{(x-y)(xy-1)}{xy}.$$

Division by xy is allowed because both factors are nonzero. Consequently,

$$h(x) = h(y) \quad \Longleftrightarrow \quad y = x \text{ or } y = x^{-1}.$$

The elements equal to their own inverse satisfy $x^2 = 1$, so they are exactly $1, -1$. They are distinct because p is odd. Their values are also distinct: $h(1) = 2$ and $h(-1) = -2$. The remaining $p - 3$ elements split into pairs $\{x, x^{-1}\}$, each giving a value not obtained from another pair. The number of values is therefore

$$2 + \frac{p-3}{2} = \frac{p+1}{2}.$$

In $\mathbb{Z}_{11}$, the pairs and singletons are

$$\{1\},\ \{10\},\ \{2,6\},\ \{3,4\},\ \{5,9\},\ \{7,8\}.$$

Their respective values of $x + x^{-1}$ are $2, 9, 8, 7, 3, 4$, so the set of values is

$$\{2, 3, 4, 7, 8, 9\}.$$

We've got it. Victory!

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3.4 Finite fields

NOTE

For a prime p, we write $\mathbb{Z}_p = \{0, \ldots, p-1\}$ and perform addition and multiplication modulo p. These integers represent residue classes. The same field is also denoted $\mathbb{F}_p$. Most finite-field examples here use these residue fields.

Definition and notation

DEFINITION 3.3

- A **finite field** is a field with finitely many elements. Write $|\mathbb{F}| = q$.
- Every finite field has $q = p^n$ elements, where p is prime and $n \geq 1$ is an integer. This is a structural theorem about finite fields.
- For a prime p, the set $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$, with addition and multiplication modulo p, is a field. It is called the **prime field of characteristic** p.

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- Addition and multiplication in $\mathbb{Z}_p$ are defined by

$$a +_{\mathbb{Z}_p} b := (a + b) \bmod p, \quad a \times_{\mathbb{Z}_p} b := (a \cdot b) \bmod p.$$

Examples

EXAMPLE 3.4

- $\mathbb{Z}_2 = \{0, 1\}$, with $1 + 1 = 0$ and $1 \cdot 1 = 1$.
- $\mathbb{Z}_7 = \{0, 1, \ldots, 6\}$, with arithmetic modulo 7; for example, $5 + 4 = 2$ and $3 \cdot 5 = 1$ in the field.
- $\mathbb{Z}_{11} = \{0, 1, \ldots, 10\}$, with arithmetic modulo 11; for example, $8 + 7 = 4$ and $4 \cdot 3 = 1$ in the field.

Basic properties

PROPOSITION 3.4

In every finite field $\mathbb{F}_q$, the set $\mathbb{F}_q^\times$ of nonzero elements is closed under multiplication, and each of its elements has a unique inverse.

PROPOSITION 3.5

In every finite field $\mathbb{F}_q$, each $a \in \mathbb{F}_q$ satisfies

$$a^q = a.$$

PROPOSITION 3.6

Every nonzero a in a finite field $\mathbb{F}_q$ satisfies $a^{q-1} = 1$.

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Explanation: Closure follows because a field has no zero divisors. The existence and uniqueness of inverses follow from the field axioms and the proof above. To prove both power identities, let $a \neq 0$ and put

$$P = \prod_{x \in \mathbb{F}_q^\times} x \neq 0.$$

Multiplication by a permutes the nonzero elements: the map $x \mapsto ax$ is invertible, with inverse $x \mapsto a^{-1}x$. Thus

$$P = \prod_{x \in \mathbb{F}_q^\times} (ax) = a^{q-1}P.$$

Since $P \neq 0$, cancellation is permitted and gives $a^{q-1} = 1$, hence $a^q = a$. For $a = 0$, the identity $0^q = 0$ is immediate.

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EXAMPLE 3.5

The nonzero inverses in $\mathbb{Z}_{11}$ are

a	1	2	3	4	5	6	7	8	9	10
a^{-1}	1	6	4	3	9	2	8	7	5	10

For example, $2 \cdot 6 = 12 \equiv 1$, $5 \cdot 9 = 45 \equiv 1$ and $7 \cdot 8 = 56 \equiv 1 \pmod{11}$.

EXERCISE 3.7

Please show that, in every finite field $\mathbb{F}_q$,

$$(a-b)^q = a^q - b^q.$$

WORKED SOLUTION

Use the identity $x^q = x$, proved for every $x \in \mathbb{F}_q$. In particular, it applies to $a - b$, so

$$(a-b)^q = a - b = a^q - b^q.$$

This proves the claim directly, without calculating binomial coefficients.

That does it. Victory!

EXERCISE 3.8

Please work in $\mathbb{Z}_{17}$.

1. Please find the inverses of $2, 3, 5, 7, 8$.
2. Please calculate $E = (7^{-1} + 3)(5 - 2^{-1})^{-1} - 2$.

WORKED SOLUTION

All arithmetic is modulo 17. We have

$$2 \cdot 9 = 18 \equiv 1, \quad 3 \cdot 6 = 18 \equiv 1, \quad 5 \cdot 7 = 35 \equiv 1,$$

$$7 \cdot 5 = 35 \equiv 1, \qquad 8 \cdot 15 = 120 \equiv 1.$$

Thus the inverses, in order, are $9, 6, 7, 5, 15$. Now $7^{-1} + 3 = 5 + 3 = 8$, while

$$5 - 2^{-1} = 5 - 9 = -4 \equiv 13 \neq 0.$$

The inverse of 13 is 4, because $13 \cdot 4 = 52 \equiv 1$. Therefore

$$E = 8 \cdot 4 - 2 = 30 \equiv 13 \pmod{17}.$$

All wrapped up. Victory!

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EXERCISE 3.9

Please show that for every nonzero a in a finite field $\mathbb{F}_q$, there is $n \in \mathbb{N}$ with $a^n = 1$.

WORKED SOLUTION

The set $\mathbb{F}_q^\times$ contains exactly $q-1$ elements and is closed under multiplication. Put $m = q-1$ and consider the list

$$a, a^2, a^3, \ldots, a^{m+1}.$$

It has $m+1$ entries but only m possible distinct values, so two entries are equal, say $a^k = a^l$ with $k > l$. Multiplying by a^{-l} gives $a^{k-l} = 1$. Thus every nonzero element has a positive integer power equal to the multiplicative identity.

We've reached the finish. Victory!

3.5 Calculations and applications in finite fields

EXERCISE 3.10

Let $a, b, c, d \in \mathbb{Z}_2$ be four bits. Add a bit e such that $a+b+c+d+e = 0$. Please find e and please calculate it for $(a, b, c, d) = (1, 0, 1, 1)$. Please determine whether the check detects one flipped bit and what happens with two flipped bits.

WORKED SOLUTION

In $\mathbb{Z}_2$, $-x = x$, so $e = a+b+c+d$. For the given example, $e = 1+0+1+1 = 1$. Write the five error bits as $\varepsilon_1, \ldots, \varepsilon_5$, with 1 indicating a flip and 0 no flip. The received parity is

$$(a+b+c+d+e) + \sum_{j=1}^{5} \varepsilon_j = \sum_{j=1}^{5} \varepsilon_j.$$

A single flip gives parity 1 and is detected. Two flips give $1+1 = 0$ and are not detected. The check does not identify which bit flipped.

Yes! Victory!

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Additional practice

EXERCISE 3.11 FOUNDATION

In $\mathbb{Z}_{11}$, define

$$A = \{x \in \mathbb{Z}_{11} : 3x + 5 = 2\}, \qquad B = \{x \in \mathbb{Z}_{11} : x^2 = 1\}.$$

Please find $A, B, A \cup B, A \cap B$ and please determine whether $A \subseteq B$.

WORKED SOLUTION

All arithmetic is modulo 11. The first equation gives $3x = 8$. Since $3^{-1} = 4$, we get $x = 4 \cdot 8 = 32 = 10$, so $A = \{10\}$. The second equation is equivalent to $(x-1)(x+1) = 0$. A field has no

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EXERCISE 3.11 FOUNDATION CONTINUED

zero divisors, so $x = 1$ or $x = -1 = 10$. Hence

$$B = \{1, 10\}, \quad A \subseteq B, \quad A \cup B = \{1, 10\}, \quad A \cap B = \{10\}.$$

V-I-C-T-O-R-Y!

EXERCISE 3.12 CHALLENGE

Let F be a finite field with $q > 2$ elements. Please prove that

$$\sum_{x \in F} x = 0, \qquad \sum_{x \in F^\times} x^{-1} = 0.$$

Please check both statements in the field with two elements. Do not assume an ordering on F.

WORKED SOLUTION

Put $S = \sum_{x \in F} x$. Since $q > 2$, choose $c \in F \setminus \{0, 1\}$. The map $x \mapsto cx$ is bijective, with inverse $x \mapsto c^{-1}x$. It therefore only rearranges the summands, and

$$cS = \sum_{x \in F} cx = \sum_{x \in F} x = S.$$

Thus $(c - 1)S = 0$. Since $c - 1 \neq 0$, field cancellation gives $S = 0$. Removing the summand 0 does not change the sum, so $\sum_{x \in F^\times} x = 0$. The map $x \mapsto x^{-1}$ also permutes $F^\times$, because applying it twice returns x. Therefore

$$\sum_{x \in F^\times} x^{-1} = \sum_{x \in F^\times} x = 0.$$

In $\mathbb{Z}_2$, we have $0 + 1 = 1 \neq 0$, and the sum of the nonzero inverses is $1^{-1} = 1 \neq 0$. Indeed, no element c distinct from $0, 1$ exists there, so the choice used in the proof is unavailable.

Victory!

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EXERCISE 3.13 CHALLENGE

Let F be a field satisfying $x^3 = x$ for every $x \in F$. Please prove that F has either two or three elements. Please write its operations in each case and please identify them with those of $\mathbb{Z}_2$ or $\mathbb{Z}_3$.

WORKED SOLUTION

For $x \neq 0$, cancellation gives $x^2 = 1$, hence $(x-1)(x+1) = 0$. A field has no zero divisors, so $x = 1$ or $x = -1$. Thus $F \subseteq \{0, 1, -1\}$. If $1 = -1$, then $F = \{0, 1\}$ and $1 + 1 = 0$. Together with multiplication by zero and $1 \cdot 1 = 1$, these are the operations of $\mathbb{Z}_2$. If $1 \neq -1$, all three elements are distinct and present. The sum $1 + 1$ cannot be 0, which would give $1 = -1$, or 1, which would give $1 = 0$ by cancellation. Therefore $1 + 1 = -1$. Writing $2 = -1$, we obtain

$$1 + 2 = 0, \qquad 2 + 2 = 1, \qquad 2 \cdot 2 = 1.$$

These, together with the zero and identity rules, give exactly $\mathbb{Z}_3$. Conversely, direct substitution of $0, 1$ in $\mathbb{Z}_2$, and $0, 1, -1$ in $\mathbb{Z}_3$, verifies $x^3 = x$.

And we're done. Victory!

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EXERCISE 3.14 CHALLENGE

Let F be a finite field. Please prove

$$\prod_{x \in F \setminus \{0\}} x = -1.$$

Here $\prod$ multiplies all nonzero elements. Please also prove that, for every $b \in F$,

$$\prod_{a \in F \setminus \{b\}} (b - a) = -1.$$

WORKED SOLUTION

Pair each nonzero x with x^{-1}, except when the two are equal. Each pair contributes 1 to the product. The unpaired elements satisfy $x = x^{-1}$, equivalently

$$x^2 = 1, \qquad (x-1)(x+1) = 0.$$

Since a field has no zero divisors, these are exactly 1 and -1. If they are distinct, their product is -1. If $1 = -1$, only the element 1 remains unpaired, and its value is again -1. This proves the first identity in every characteristic. For fixed b, the values $b - a$ as a ranges over $F \setminus \{b\}$ run through every nonzero field element exactly once. The second product is therefore the same product in a different order.

You know what that means? V-I-C-T-O-R-Y!

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